

Topic Modelling of Donald Trump Rally Speeches Using BERTopic

Introduction

This project implements an unsupervised machine learning pipeline to analyze a text corpus of Donald J. Trump’s campaign rally transcripts from 2019 and 2020. Political comitia represent a challenging natural language processing (NLP) domain, characterized by highly unstructured prose, repetitive rhetorical formulas, and rapid shifts between policy statements, adversarial attacks, and local references. The core objective is twofold: to isolate latent thematic pillars within this complex discourse and to systematically evaluate their distribution across chronological and geographical dimensions.

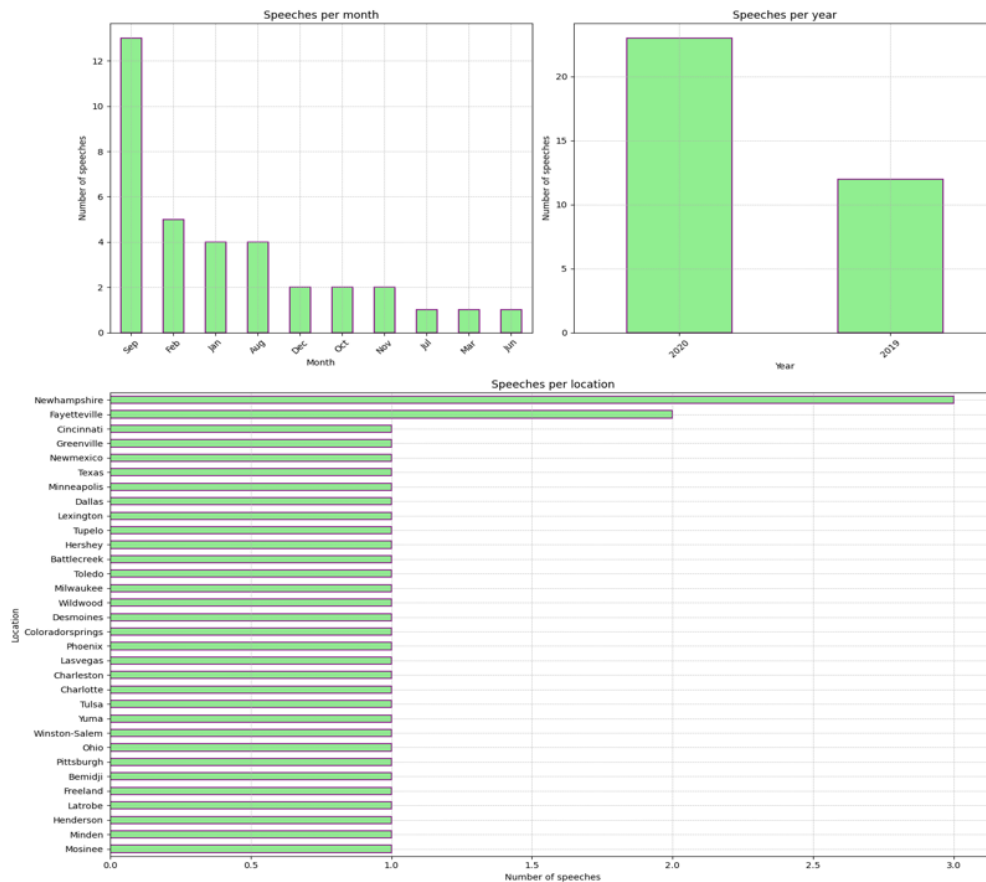
Dataset

The selected dataset contains 35 speeches by Trump from 2019 and 2020. Each filename contains useful metadata, such as the location, month, day, and year of the speech. These elements were extracted using regular expressions.

The final dataset includes the following main variables:

- “filename”: name of the original text file;
- “text”: full speech transcript;
- “year”: year of the speech;
- “month”: month of the speech;
- “location”: location extracted from the filename.

As illustrated in the figure, the corpus exhibits a significant temporal skew toward 2020, peaking dramatically in September due to the intensification of the presidential campaign cycle. Geographically, while the layout is highly distributed, strategic battleground locations—such as New Hampshire—register higher event density.

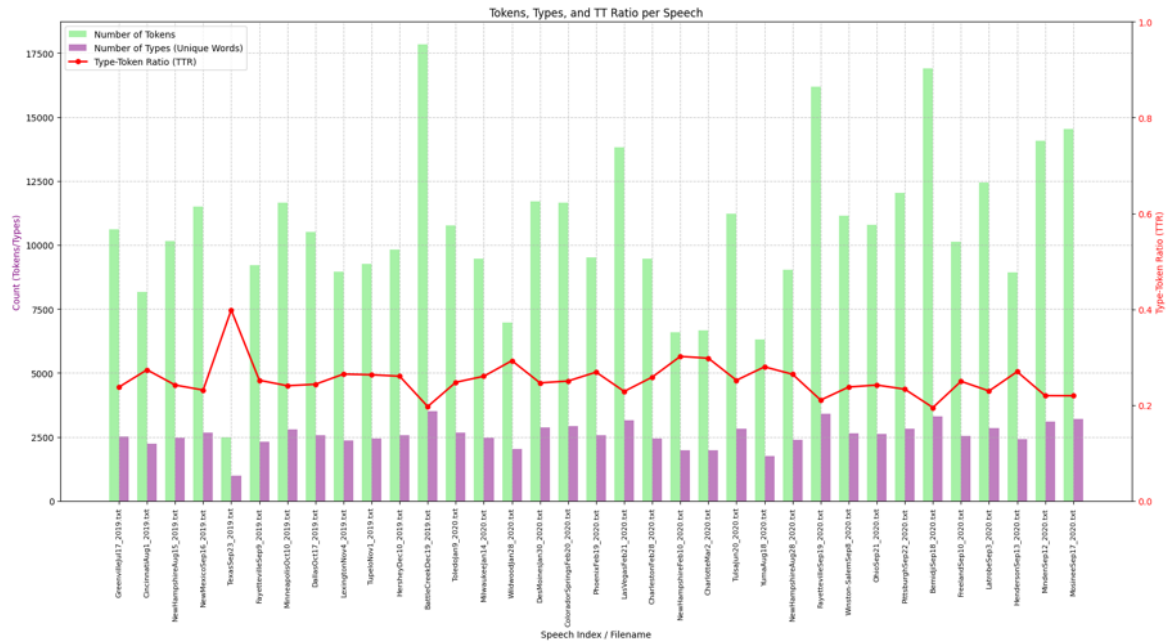


Exploratory Lexical Analysis

Prior to thematic modeling, statistical properties of the text were computed to assess linguistic density and structural predictability. The initial baseline tracks total word count (tokens), vocabulary breadth (types), Type-Token Ratio (TTR), and Shannon Entropy across all documents.

The corpus displays extreme structural variance: individual transcripts range from a baseline of 2,500 tokens to exceeding 18,000. This significant discrepancy introduces a methodological challenge. Longer political addresses naturally cycle through multiple distinct sub-topics, increasing semantic dilution and potentially degrading the performance of standard document-level topic modeling.

The type-token ratio was used as a basic measure of lexical diversity. TTR exhibits a strict inverse correlation with document length. Shorter speeches yield a higher TTR, whereas expansive addresses experience a sharp decline in lexical density. This trend indicates that prolonged comitia do not introduce a wider vocabulary, but rather rely on the continuous repetition of a highly consolidated core lexicon. Concurrently, calculated Shannon Information Entropy values suggest low structural variance, confirming a highly predictable statistical structure behind the speaker's conversational style.



Preprocessing

To mitigate linguistic noise and standardize the text data for dense vector embedding, a programmatic preprocessing pipeline was executed using the NLTK framework. The data preparation follows a rigorous five-stage pipeline:

1. **Tokenization:** Text sequences were segmented into discrete lexical tokens using the NLTK word tokenizer, cleanly separating punctuation and isolating standard English verbal contractions (e.g., *"don't"* mapped to *"do"* and *"n't"*);
2. **Punctuation and Symbol Elimination:** All non-alphanumeric characters, punctuation marks, and structural formatting artifacts were programmatically removed;
3. **Case Normalization:** Universal lowercasing was applied across the text matrix to eliminate duplicate token identities caused by capitalization;
4. **Stopword Filtering and Noise Removal:** Standard NLTK stopwords lists were utilized to strip out high-frequency functional words. This was augmented by a custom domain mask to filter out repetitive conversational filler specific to rally scripts (e.g., *'ve*, *'re*, *'s*, *lol*);
5. **Numerical Stripping:** All numerical tokens were discarded to ensure the subsequent embedding layers focus exclusively on qualitative semantic units.

Chunking

A central methodological choice was the division of speeches into smaller chunks. This step was necessary because a full rally speech usually contains multiple themes, such as the economy, immigration, foreign policy, political opponents, media criticism, and campaign slogans. If an entire speech were treated as a single document, its embedding would represent an average of all these themes, reducing topic interpretability.

Each speech was partitioned into uniform chunks of 200 tokens, maintaining an overlap window of 50 tokens to ensure narrative and semantic continuity across boundaries. This operation expanded the initial 35 multi-topic texts into a matrix of 1,158 semantic units. Each derived chunk retains its inherited metadata coordinates (chronological identifiers, geopolitical location, and positional index). Baseline tests executing topic modeling directly on unfragmented speeches confirmed the necessity of this step: without chunking, data points suffered from extreme semantic overlap, preventing the isolation of distinct, interpretatively relevant clusters.

Topic Modelling

After preparing the data, topic modelling techniques were applied to identify the main themes contained in the speeches. The main goal was to use BERTopic, but before applying the library directly, the main steps behind its functioning were manually implemented. This was fundamental to understand the pipeline more clearly and test the effect of different parameters.

The manual pipeline included the following steps:

1. **Word Embedding with SBert**

Each chunk was transformed into a dense semantic embedding and mapped into a continuous vector space. Unlike traditional sparse bag-of-words arrays, these dense representations capture deep latent contextual relationships, outputting high-dimensional embedding vectors. The resulting embeddings had 384 dimensions.

2. **Dimensionality Reduction with UMAP**

UMAP was used to compress the embedding vectors from 384 down to 30 dimensions, maximizing local semantic structure retention while optimizing downstream computational efficiency.

3. **Clustering with HDBSCAN**

This algorithm groups vector topologies into natural, density-defined shapes, while automatically segregating loose conversational filler or ambiguous segments as statistical noise outliers (assigned to Cluster -1).

4. **Final Preparation of data with Count Vectorizer**

A CountVectorizer was deployed to convert clustered text segment matrices into structural token frequency representations. To guarantee high-quality topic indicators, a customized stopword list was integrated to target domain-specific, non-informative frequent words (e.g., *"thanks"*, *"everybody"*, *"hello"*). Crucially, this specialized vocabulary filtering was purposely deferred to this downstream topic representation stage rather than being executed during early preprocessing. Because Sentence-BERT rely on full syntactic context to extract precise semantic meaning, preserving these conversational elements during embedding generation prevented performance degradation, isolating the noise reduction strictly to the keyword extraction phase.

5. Topic Representation with C-TF-IDF

After clustering, c-TF-IDF was used to extract characteristic keywords for each isolated theme. Unlike the standard TF-IDF architecture, which evaluates terms across individual standalone text entries, c-TF-IDF treats all text chunks assigned to a single HDBSCAN cluster as a single concatenated macro-document. By mathematically comparing token distributions within an isolated cluster against their global corpus frequencies, the algorithm successfully extracts highly representative, distinctive keyword sets that maximize semantic interpretability.

6. Evaluation through Coherence and Diversity

Given the unsupervised nature of the task, the custom architecture was quantitatively evaluated using semantic Coherence and Topic Diversity metrics rather than traditional accuracy markers. Coherence measures the semantic relatedness of the top predicted keywords, while Diversity assesses the degree of thematic distinction between the generated categories.

The custom modular pipeline achieved a coherence score of approximately 0.65 and a topic diversity score of approximately 0.8, demonstrating a high degree of interpretability and low semantic overlap.

To validate our custom modular setup, the multi-stage manual workflow was compared to the BERTopic library. The end-to-end library natively unifies embedding generation, UMAP manifold reduction, HDBSCAN partitioning, and c-TF-IDF extraction, confirming the architectural consistency and technical robustness of our custom pipeline.

Deploying this automated library enabled the efficient extraction of structured final topics, diagnostic metadata, and representative text documents. Furthermore, it showed advanced analytical features, including semantic similarity matrices, hierarchical thematic charts, and multi-dimensional tracking of topic evolution across both chronological and geographical coordinates.

In summary, while the manual implementation was crucial for isolating and verifying the core mathematical logic at each computational step, the integrated BERTopic library served as the optimal deployment engine to scale the corpus and generate highly structured interpretive insights.

Interpretation of Results

The analysis shows that Trump's rally speeches are highly repetitive but also thematically dense. Many speeches combine several recurring political themes, including economic success, immigration and border security, criticism of Democrats, criticism of the media, foreign policy, China, law and order, and campaign mobilization.

The chunking strategy proved useful because it allowed the model to detect more specific themes than a document-level approach would have allowed. Instead of assigning one broad topic to an entire speech, the model could identify smaller thematic segments within each rally.

Since BERTopic relies on dimensionality reduction and density-based clustering, slight variations may occur across different runs if no random seed is fixed. In particular, UMAP introduces

stochasticity during the projection of sentence embeddings into a lower-dimensional space, which can affect the subsequent HDBSCAN clustering step and, consequently, the final topic representations. To ensure reproducibility, a fixed random state was set for UMAP and the main random seeds were initialized before training the model. However, small changes in topic composition and topic labels may still occur.

Despite these variations, the main thematic areas remained relatively stable across different runs. The recurring themes include a large outlier/noise category, immigration and border security, especially connected to Hispanic voters, the wall, and sanctuary cities; criticism of Biden, the left, and impeachment; foreign policy references to Israel and Jerusalem; national security narratives connected to Soleimani and al-Baghdadi; healthcare and drug prices; and local political references, especially to Michigan, North Carolina, and South Carolina. This suggests that, even if the exact topic structure may change slightly, the core rhetorical patterns of the corpus are robust.

The topics, ordered from the largest to the smallest, are the following.

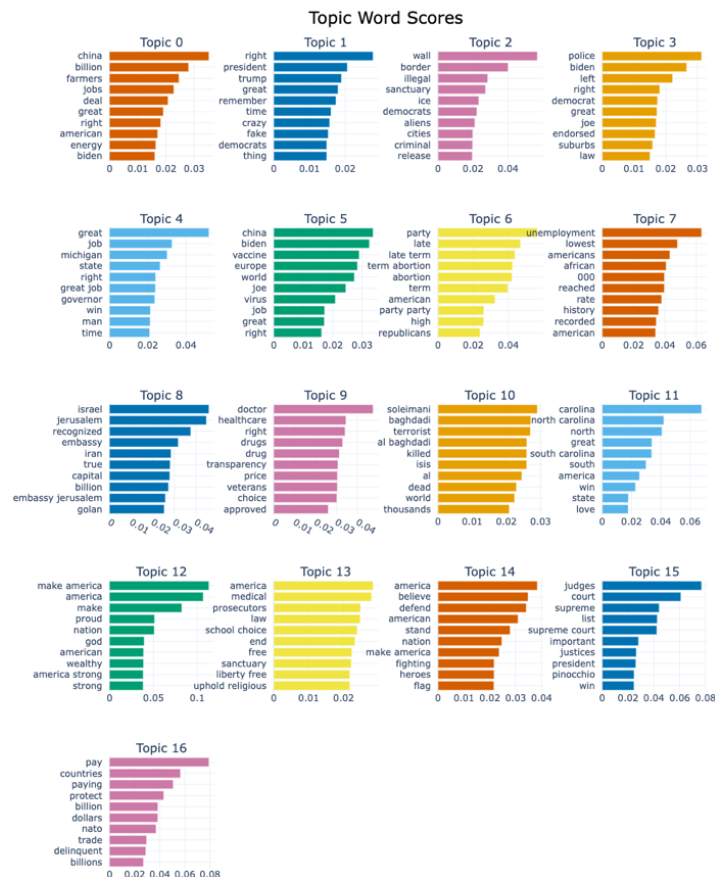
- a. “-1 – Outliers”: this topic includes all those chunks that HDBSCAN couldn’t assign to any cluster. It contains very general and repetitive words such as “great”, “right”, “president”. This means that many passages of speeches are made up of recurring formulas, transitional phrases, slogans or expressions that are too generic to create a semantically strong cluster. It shows the repetitive and performative nature of rallies.
- b. “0 - China, farmers, jobs, trade”: is the biggest topic, composed of 150 chunks; the main words are “china, billion, farmers, jobs, deal, american, billions”. Trump repeatedly frames trade policy, especially toward China, as a battle to protect American farmers, workers, and jobs.
- c. “1 – Trump, president, general campaign rhetoric”: this topic includes 126 chunks. Its main words are “right”, “president”, “trump”, “great”, and “remember”. This topic is less issue-specific and contains general campaign rhetoric, self-referential statements, and repeated evaluative language. It is close to the noise category, but it still reflects a recognizable rhetorical pattern centered on Trump’s personal and political narrative.
- d. “2 - Wall, border, sanctuary cities, illegal immigration”: composed of 100 chunks; the main words are “wall, border, sanctuary, illegal, ice, aliens, cities, democrats”. This topic shows that immigration is framed through the language of national security, border control, criminality, and opposition to Democratic sanctuary policies. It is one of the clearest and most interpretable topics in the model.
- e. “3 - Police, Biden, left, law and order”: composed of 82 chunks; the main words are “police, biden, left, joe, suburbs, law, Portland”. This topic reflects Trump’s law-and-order rhetoric, where Biden and the political left are associated with disorder, crime, protests, and threats to suburban safety. It is especially relevant to the 2020 campaign context.
- f. “4 - Michigan, state, local campaign references”: composed of 77 chunks; the main words are “great, job, michigan, state, governor, win”. This topic captures local campaign

references and state-specific political appeals, especially connected to Michigan and electoral mobilization. It shows that some topics are not purely national issues, but are tied to specific campaign locations and audiences.

- g. “5 - Biden, China, Europe, vaccine, virus”: composed of 44 chunks; the main words are “biden, china, europe, vaccine, world, virus”. This topic reflects the intersection between the 2020 campaign, COVID-related discourse, China, and criticism of Biden. It shows how pandemic-related issues became part of the campaign narrative.
- h. “6 - Abortion, party, late-term abortion”: composed 37 chunks; the main words are “party, abortion, late, late term, republicans, American”. This topic captures moral and cultural issues, especially abortion and the contrast between Republican and Democratic values. It belongs to the broader category of culture-war rhetoric.
- i. “7 - Unemployment, economy, minority groups”: composed of 29 chunks; the main words are “unemployment, lowest, reached, rate, history, americans, african, American”. This topic reflects Trump’s economic achievement narrative, especially claims about historically low unemployment rates among different demographic groups
- j. “8 - Israel, Jerusalem, Iran, foreign policy”: composed of 28 chunks; the main words are “israel, jerusalem, recognized, iran, billion, deal”. This topic captures Trump’s foreign policy discourse, especially references to Israel, Jerusalem, Iran, and international agreements. It is more specific than topics such as economy or immigration, and therefore appears in fewer chunks.
- k. “9 - Healthcare, drugs, doctors”: composed 28 chunks; the main words are “doctor, healthcare, drugs, drug”. This topic captures healthcare-related discourse, including drug prices, medical treatment, and health policy achievements.
- l. “10 - Soleimani, ISIS, terrorism, military action”: composed of 22 chunks; the main words are “soleimani, isis, killed, terrorist, albaghdadi, dead, military”. This topic reflects Trump’s national security narrative, focused on military strength, counterterrorism, and the elimination of foreign enemies.
- m. “11 - Carolina, local speeches, regional references”: composed of 21 chunks; the main words are “carolina, north carolina, south carolina, state, win, open”. This topic captures geographically specific campaign language, especially references to North Carolina and South Carolina. It shows that the model also detects location-based rhetorical patterns.
- n. “12 - America, patriotism, national identity”: composed of 17 chunks; the main words are “america, make america, american, nation, defend, god, love”. The topic shows how Trump constructs a patriotic narrative based on national pride, religious references, and the idea of

defending America. It is closely connected to the broader “Make America Great Again” rhetorical frame.

- o. “13 - Medical system, Law”: composed of 19 chunks; the main words are “america, law, medical, end, defend, prosecutor”. Examines institutional governance, centering on legal protection frameworks, medical system defenses, and the rule of law.
- p. “14 – America, belief, defense, national values”: this topic includes 15 chunks. Its main words are “america”, “believe”, “defend”, “american”, and “stand”. It captures another patriotic and value-based rhetorical cluster, focused on defending America, national values, and collective political identity.
- q. “15 - Judges, Supreme Court, Second Amendment”: composed of 14 chunks; the main words are “judges, court, supreme, supreme court, second amendment, justices”. This topic captures Trump’s emphasis on judicial appointments, the Supreme Court, constitutional rights, and the Second Amendment.
- r. “16 – Foreign countries, payments, protection”: this topic includes 12 chunks. Its main words are “pay”, “countries”, “paying”, “protect”, and “billion”. It reflects Trump’s discourse on foreign allies, defense spending, and the idea that other countries should pay more for American protection.

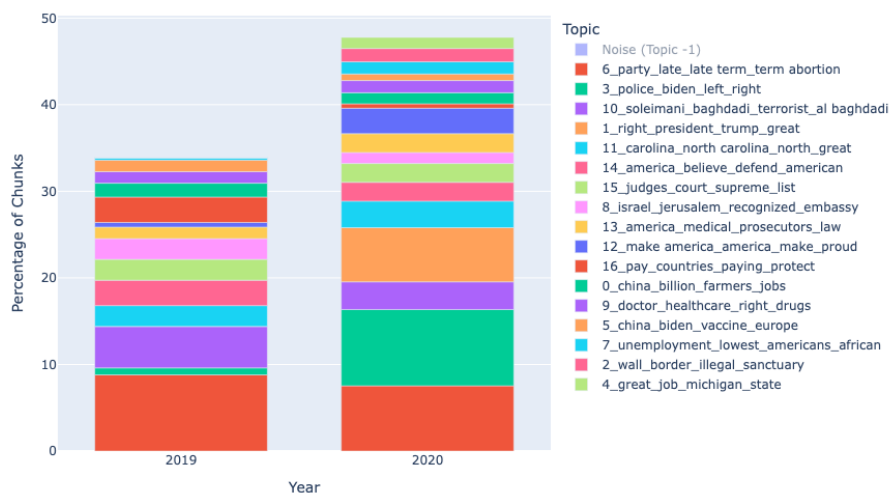


To further deepen the research, the topics were assessed over their yearly, monthly and spatial evolution. Also, their semantic similarity was studied.

i. Topic evolution over the years

The yearly distribution of topics shows a clear thematic shift between 2019 and 2020. In both years, the noise topic remains the most frequent category, indicating that a large portion of the corpus consists of repetitive, mixed, or rhetorically fragmented speech segments that are difficult to assign to a single coherent topic. However, the comparison between the two years also reveals a stronger presence of adversarial and campaign-oriented themes in 2020. Topics related to Biden, the left, police, impeachment, and the Democratic Party become more prominent, suggesting an intensification of partisan conflict during the presidential election cycle. At the same time, economic topics such as China, farmers, jobs, and unemployment remain relatively stable across the two years. The Israel-related topic, associated with terms such as Israel, Jerusalem, and recognition, also appears more clearly in 2020, indicating the presence of a foreign-policy narrative connected to diplomatic achievements and international positioning. Overall, the yearly analysis suggests that Trump's discourse combines a stable ideological core, mainly based on economic achievement and trade protectionism, with more context-dependent themes linked to electoral conflict, foreign policy, and the specific political dynamics of the 2020 campaign.

Topic Evolution Over Years (Percentage of Chunks per Topic per Year)

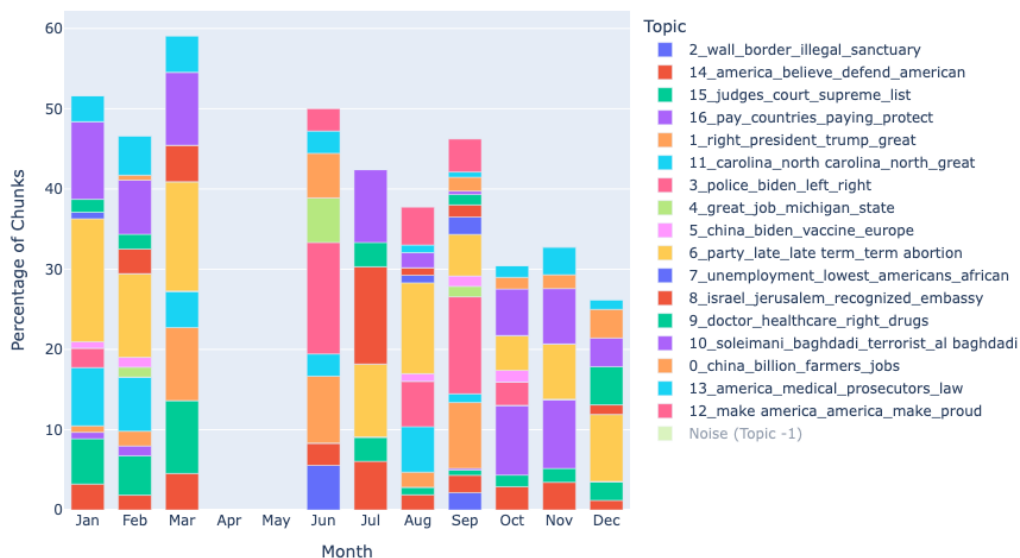


i. Topics over months

The monthly distribution of topics provides a more detailed view of the temporal variation within the corpus. The noise topic remains consistently high across most months, confirming the presence of repetitive and mixed speech segments. However, some topics show clear monthly concentrations. The law-and-order topic, associated with police, Biden, and the left, becomes more visible during the central months of the

2020 campaign, suggesting a stronger emphasis on crime, protests, and public order in the summer and early autumn. The topic related to Soleimani, ISIS, terrorism, and military action is concentrated especially at the beginning of the year. This pattern is coherent with the fact that Qasem Soleimani was killed in January 2020 on Trump's order, making military success and national security particularly salient in that period. In contrast, economic themes such as unemployment, China, farmers, jobs, and trade appear more regularly across different months. Overall, the monthly analysis shows that Trump's discourse combines stable economic narratives with more event-driven topics linked to foreign policy, law and order, and campaign conflict.

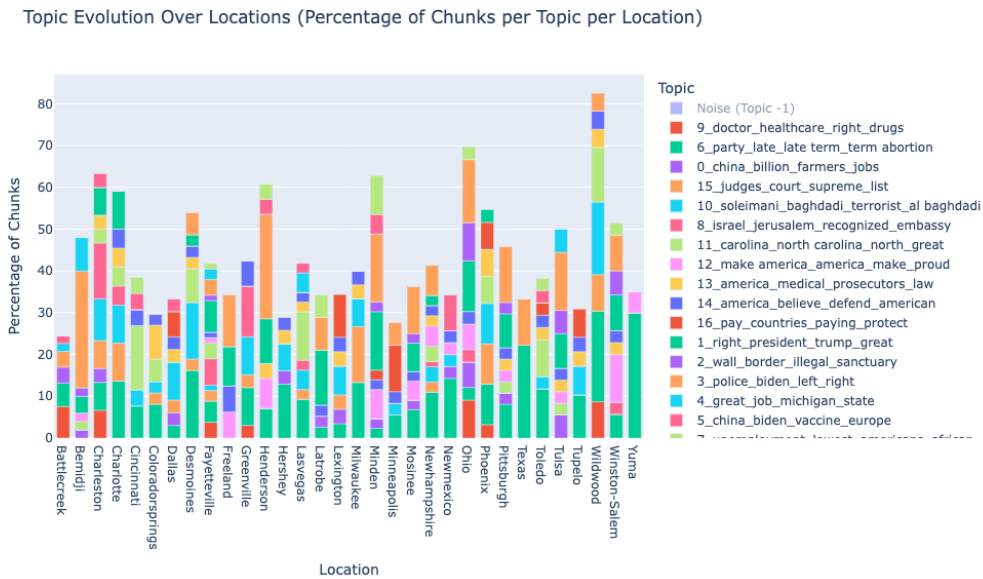
Topic Evolution Over Months (Percentage of Chunks per Topic per Month)



ii. Topics over locations

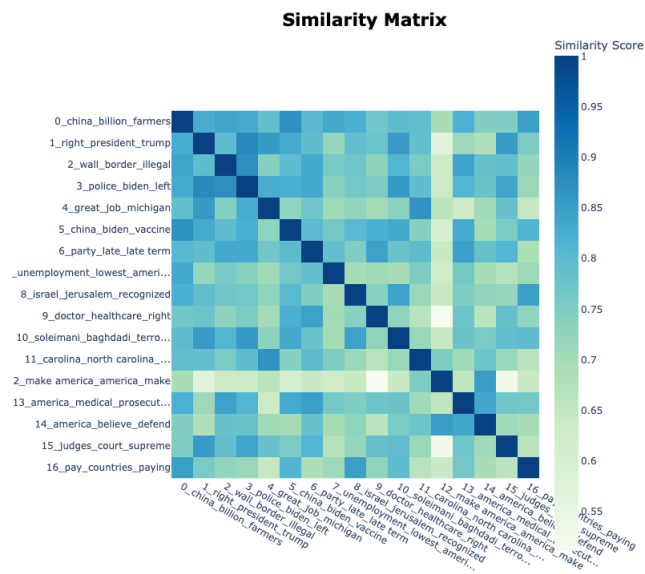
The distribution of topics across locations shows that Trump's speeches combine a stable national repertoire with local rhetorical adaptation. The noise topic remains high in most locations, confirming the presence of repetitive and mixed speech segments across the corpus. However, some locations display stronger concentrations of specific themes. For example, Henderson, Minden, Pittsburgh, and Tulsa show a more visible presence of the law-and-order topic, suggesting that references to police, Biden, the left, crime, and public order were emphasized more strongly in these rallies. Other locations show a greater presence of healthcare-related discourse, indicating that issues such as doctors, healthcare, rights, and drug prices were selectively emphasized rather than uniformly distributed. Border-related and Hispanic-related topics also appear more strongly in specific locations, especially where immigration and border security were particularly relevant to the local campaign context. Overall, the location analysis suggests that Trump's rhetoric relies on recurring national themes, but adjusts their relative

importance according to the audience and the political context of each rally.



iii. **Similarity between topics**

The similarity matrix shows that the extracted topics are semantically coherent but not redundant. Topics belonging to the same broader area, such as border security, immigration, and Hispanic-related discourse, appear closer to one another, indicating that the model captures meaningful thematic connections. Similarly, topics related to Biden, Democrats, impeachment, police, and the left share a common conflict-oriented structure. At the same time, more distinct themes, such as healthcare, unemployment, judges, and foreign policy, remain comparatively separated. Overall, the matrix supports the validity of the model, showing that the topics are related within the same political discourse but still represent different narrative areas.



Limitations

This project has some limitations. First, the corpus is relatively small and limited to a specific set of Trump rally speeches from 2019 and 2020, so the results cannot be generalized to Trump's entire political communication. Second, the analysis depends on preprocessing choices, such as stopword removal, chunk size, and custom Trump-related stopwords, which can influence the final topic structure. Third, BERTopic is not fully deterministic: although a fixed seed and random state were used to improve reproducibility, small variations in the number, size, and labels of topics may still occur. Finally, topic interpretation requires qualitative analysis and knowledge of the historical and political context. The topic labels should therefore be understood as informed analytical descriptions, rather than objective or definitive categories.

Conclusions

This project applied a complete text mining pipeline to a corpus of Donald Trump rally speeches. The workflow included preprocessing, chunking, Sentence-BERT embeddings, UMAP dimensionality reduction, HDBSCAN clustering, and c-TF-IDF topic representation through BERTopic.

The results show that Trump's rally speeches are characterized by strong repetition, recurrent thematic structures, and frequent topic switching. The main themes concern trade, China, jobs, immigration, border security, law and order, patriotism, healthcare, foreign policy, and electoral attacks against political opponents.

The most relevant methodological choice was the use of chunking, which allowed the model to capture multiple themes within the same speech. Overall, BERTopic proved to be an effective exploratory framework for identifying recurrent political narratives and analysing their variation across time and location.